19 The point P has coordinates (3, 4)
The point Q has coordinates (a, b)

A line perpendicular to PQ is given by the equation $3x + 2y = 7$

Find an expression for b in terms of a.

$3x + 2y = 7$

$-3x \qquad -3x$

$2y = 7 - 3x$

$\div 2 \qquad \div 2$

$y = \frac{-3}{2}x + \frac{7}{2}$

1 mark
Rearrange to identify the gradient

Perpendicular gradients multiply to -1.

So the gradient of PQ is $\frac{2}{3}$

1 mark
Use the negative reciprocal

Using P(3, 4) in $y = \frac{2}{3}x + c$:

$4 = \frac{2}{3} \times 3 + c = 2 + c$, so $c = 2$.

1 mark
Use the known point to find the intercept

For Q(a, b), substitute $x = a$ and $y = b$:

$b = \frac{2}{3}a + 2$

1 mark
Form an equation in a and b

1 mark
Correct expression for b

$b = \frac{2}{3}a + 2$

(Total for Question 19 is 5 marks)